

1. Question: Networking refers to the linking of computers and what other devices?

- A) Standalone printers
- B) Communication network devices (hosts)
- C) Memory storage units
- D) Power supply units

2. Question: What separates devices in a network according to the definition of networking?

- A) Cables
- B) Hubs
- C) Unique device identifiers (IP/MAC addresses)
- D) Routers

3. Question: Which of the following is NOT a parameter used to classify computer networks?

- A) Type of connection
- B) Physical topology
- C) Network reach
- D) Device brand

4. Question: How many types of connections are there based on the way a host communicates with other hosts?

- A) One
- B) Two
- C) Three
- D) Four

5. Question: Which type of connection is used to establish direct connections between two hosts?

- A) Point-to-multipoint
- B) Broadcast
- C) Point-to-point
- D) Multicast

6. Question: A remote control for an air conditioner is mentioned as an example of what type of connection?

- A) Point-to-multipoint
- B) Point-to-point
- C) Broadcast
- D) Anycast

7. Question: In which connection type do more than two hosts share the same link?

- A) Point-to-point
- B) Simplex
- C) Point-to-multipoint
- D) Duplex

8. Question: FDMA (frequency division multiple access) is an example of what kind of channel sharing in point-to-multipoint connections?

- A) Temporal sharing
- B) Spatial sharing
- C) Code sharing
- D) Time sharing

9. Question: TV Broadcasting is given as an example of what type of channel sharing in

point-to-multipoint connections?

- A) Spatial
- B) Frequency
- C) Temporal
- D) Code

10. Question: How many broad physical topologies are mentioned for computer networks?

- A) Two
- B) Three
- C) Four
- D) Five

11. Question: In a star topology, every host has a point-to-point link to what central device?

- A) Router
- B) Switch
- C) Central controller or hub
- D) Server

12. Question: Which of the following is listed as an advantage of a star topology?

- A) Resilient against single point of failure
- B) Traffic privacy
- C) Ease of Installation
- D) Complex connections

13. Question: What is a major disadvantage of the star topology mentioned in the text?

- A) Its costly

- B) Traffic privacy and security ensured
- C) Single Point of Failure
- D) Complex connections

14. Question: In which topology is every host connected to every other host using a dedicated link?

- A) Star
- B) Bus
- C) Mesh
- D) Ring

15. Question: What is an advantage of a mesh topology?

- A) Ease of Installation
- B) Cheap to establish
- C) Resilient against single point of failure
- D) Single Point of Failure

16. Question: Which of the following is listed as a disadvantage of a mesh topology?

- A) Ease of Fault Identification
- B) Its costly
- C) Traffic is visible to all nodes
- D) Prone to single point of failure

17. Question: A bus topology follows which type of connection principle?

- A) Point-to-point
- B) Point-to-multipoint
- C) Simplex

D) Full duplex

18. Question: In a bus topology, what serves as the primary traffic pathway between the hosts?

A) Central hub

B) Dedicated links

C) Backbone cable or bus

D) Repeaters

19. Question: What is an advantage of establishing a bus topology?

A) Resilient against single point of failure

B) Its cheap to establish such connections

C) Private broadcasting is possible

D) Ease of Fault Identification

20. Question: A disadvantage of the bus topology is:

A) Its costly

B) Complex Connections

C) Localising fault in the network is difficult

D) Prone to single point of failure

21. Question: A ring topology works on the principle of what type of connection?

A) Point-to-multipoint

B) Broadcast

C) Simplex

D) Point-to-point

22. Question: In a ring topology, each host connects to its two immediate neighbouring hosts through what devices at each host?

- A) Hubs
- B) Switches
- C) Routers
- D) Repeaters

23. Question: Which of these is an advantage of a ring topology?

- A) Single Point of Failure
- B) Ease of Installation
- C) Its costly
- D) Private broadcast is possible

24. Question: A disadvantage of a ring topology is that it is:

- A) Resilient against single point of failure
- B) Complex Connections
- C) Prone to single point of failure
- D) Hard to install

25. Question: How many broad categories are computer networks divided into based on network reachability?

- A) Two
- B) Three
- C) Four
- D) Five

26. Question: Which type of network is mostly restricted to individual usage, like within a house?

- A) LAN
- B) WAN
- C) PAN
- D) MAN

27. Question: Connected wireless headphones, laptops, and wireless mouse are good examples of devices found in which network type?

- A) LAN
- B) WAN
- C) PAN
- D) MAN

28. Question: Local Area Networks (LANs) are typically restricted to what areas?

- A) States or countries
- B) Continents
- C) Buildings, organizations, or campuses
- D) Cities

29. Question: Present-day data access rates within LANs typically range from:

- A) 1 Mbps to 10 Mbps
- B) 100 Mbps to 1000 Mbps
- C) 10 Gbps to 100 Gbps
- D) 100 Kbps to 1 Mbps

30. Question: Which of the following is a commonly used network component in a LAN?

- A) Satellite-based links
- B) Public switched telephone networks (PSTNs)
- C) Servers
- D) Transoceanic cables

31. Question: The reachability of a MAN lies between that of which two network types?

- A) PAN and LAN
- B) LAN and WAN
- C) WAN and GAN
- D) PAN and MAN

32. Question: Metropolitan Area Networks (MANs) typically connect various organizations or buildings within a given:

- A) Single building
- B) State or country
- C) City
- D) Individual household

33. Question: An Internet service provider (ISP) supplying Internet connectivity to various organizations within a city is an excellent example of what network type?

- A) LAN
- B) WAN
- C) PAN
- D) MAN

34. Question: Wide Area Networks (WANs) typically connect diverse geographic locations but are

restricted within the boundaries of a:

- A) Single building
- B) City
- C) State or country
- D) Continent

35. Question: WANs connecting two LANs or MANs may use which type of links?

- A) Ethernet cables
- B) Wi-Fi
- C) Satellite-based links
- D) Bluetooth

36. Question: What are the two most commonly accepted and used traditional layered network models mentioned?

- A) Ethernet and Wi-Fi
- B) ISO-OSI and Internet protocol suite
- C) FDMA and TDMA
- D) Hubs and Switches

37. Question: How many layers of abstraction does the ISO-OSI model partition a networked communication device into?

- A) Four
- B) Five
- C) Six
- D) Seven

38. Question: Which layer is referred to as layer 1 of the OSI model?

- A) Data link layer
- B) Physical layer
- C) Network layer
- D) Transport layer

39. Question: Which of the following is a function of the Physical layer in the OSI model?

- A) Addressing of packets
- B) End-to-end error recovery
- C) Topological layout of the network
- D) Format conversion of data

40. Question: What is the Protocol Data Unit (PDU) at the Physical layer?

- A) Data Frames
- B) Packets
- C) Segments
- D) Symbol

41. Question: The Data Link layer is layer number _____ of the OSI model.

- A) 1
- B) 2
- C) 3
- D) 4

42. Question: The Data Link layer is mainly concerned with the establishment and termination of the connection between two hosts and the detection and correction of what?

- A) Routing issues
- B) Errors during communication
- C) Congestion
- D) User authentication

43. Question: What are the two subparts of the Data Link layer mentioned in the text?

- A) IP and ARP
- B) TCP and UDP
- C) Media Access Control (MAC) and Logical Link Control (LLC)
- D) HTTP and FTP

44. Question: What is the Protocol Data Unit (PDU) at the Data Link layer?

- A) Symbol
- B) Data Frames
- C) Packets
- D) Data

45. Question: Which layer of the OSI model provides a means of routing data to various hosts connected to different networks through logical paths?

- A) Physical layer
- B) Data link layer
- C) Network layer
- D) Transport layer

46. Question: What is the Protocol Data Unit (PDU) at the Network layer?

- A) Data Frames

- B) Packets
- C) Segments
- D) Data

47. Question: The Transport layer is layer number _____ of the OSI model.

- A) 2
- B) 3
- C) 4
- D) 5

48. Question: Which of the following is a function of the Transport layer?

- A) Topological layout of the network
- B) Addressing and sequencing of packets
- C) Keeping track of acknowledgements of segments
- D) Encryption of data

49. Question: Which of the following are protocols specifically mentioned at the Transport layer?

- A) IP and ARP
- B) HTTP and FTP
- C) PPP and Frame relay
- D) TCP and UDP

50. Question: Which layer is responsible for establishing, controlling, and terminating communication between networked hosts?

- A) Transport layer
- B) Session layer

C) Presentation layer

D) Application layer

---SUMMARY---

Networking involves linking computers and communication devices through unique identifiers like IP and MAC addresses across networks like the Internet or Intranet. Networks are classified by their connection type, either point-to-point or point-to-multipoint, their physical topology, such as star, mesh, bus, or ring configurations, and their geographical reach, categorized as Personal Area Networks (PAN), Local Area Networks (LAN), Metropolitan Area Networks (MAN), or Wide Area Networks (WAN). Communication within these diverse networks is governed by layered architectural models, notably the seven-layer OSI model and the four-layer Internet Protocol (TCP/IP) suite, which abstract specific tasks from the physical transmission of data to user-facing application services.